okay teach me what is threshold and why its needed in our project and also explain the code used in it

Threshold = **cutoff value to convert probability → final decision (0 or 1)**.

1. What is threshold

Model gives:

```
P(default) = 0.42
```

Threshold rule:

```
if P ≥ threshold → default (1)
else → safe (0)
```

2. Default threshold

```
threshold = 0.5
```

Means:

- ≥ 50% risk → default
- < 50% → safe

3. Why threshold is needed

Model only gives **probability**, not decision.

Without threshold:

```
0.42 → ??? (no decision)
```

Threshold converts:

```
0.42 → safe (if threshold = 0.5)
0.42 → default (if threshold = 0.4)
```

4. Why we changed threshold in YOUR project

Your goal:

```
catch defaulters (reduce FN)
```

At 0.5:

- many risky customers ignored

So we lowered threshold:

```
0.5 → 0.4
```

Now:

- moderate risk also flagged
- recall ↑

5. Tradeoff

Threshold	Effect
High (0.5)	safe, fewer false alarms
Low (0.4)	risk-sensitive, catches more defaulters

6. Code explanation

Step 1. Get probabilities

Python Run

```python
y_prob = model.predict_proba(X_test_scaled)[:,1]
```

- gives probability of default
- `[:,1]` → class 1 (default)

Step 2. Apply threshold

Python Run

```python
threshold = 0.4
y_pred = (y_prob >= threshold).astype(int)
```

Breakdown

```
y_prob = [0.3, 0.8, 0.45]
```

Step 2. Apply threshold

Python Run

```python
threshold = 0.4
y_pred = (y_prob >= threshold).astype(int)
```

Breakdown

```
y_prob = [0.3, 0.8, 0.45]
```

Check:

```
0.3 >= 0.4 → False → 0
0.8 >= 0.4 → True → 1
0.45 >= 0.4 → True → 1
```

Final:

```
[0, 1, 1]
```

Step 3. Evaluate

Python Run

```python
recall_score(y_test, y_pred)
f1_score(y_test, y_pred)
```

Check:

- how many defaulters caught

Python Run

Check:

- how many defaulters caught
- overall balance

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7. Why we tested multiple thresholds

Python Run

```python
for t in [0.3, 0.35, 0.4, 0.45]:
```

Reason:

```
find best tradeoff between recall and F1
```

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8. Big concept

```
Model training → learns probabilities
Threshold → business decision layer
```

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9. Real-world meaning

- 0.5 → only high risk flagged
- 0.4 → moderate risk also flagged

👉 safer decision-making